# LLMs + KLAS

## The Case for Compact AI

Knowledge-Level Agent Systems
for Software Engineering

Tim Menzies · NC State · timm@ieee.org

*"Obtaining state-of-the-art results can be achieved with smarter questioning, not planetary-scale computation."*

— CACM, Sept 2025

# Motivation: AI design needs a rethink?

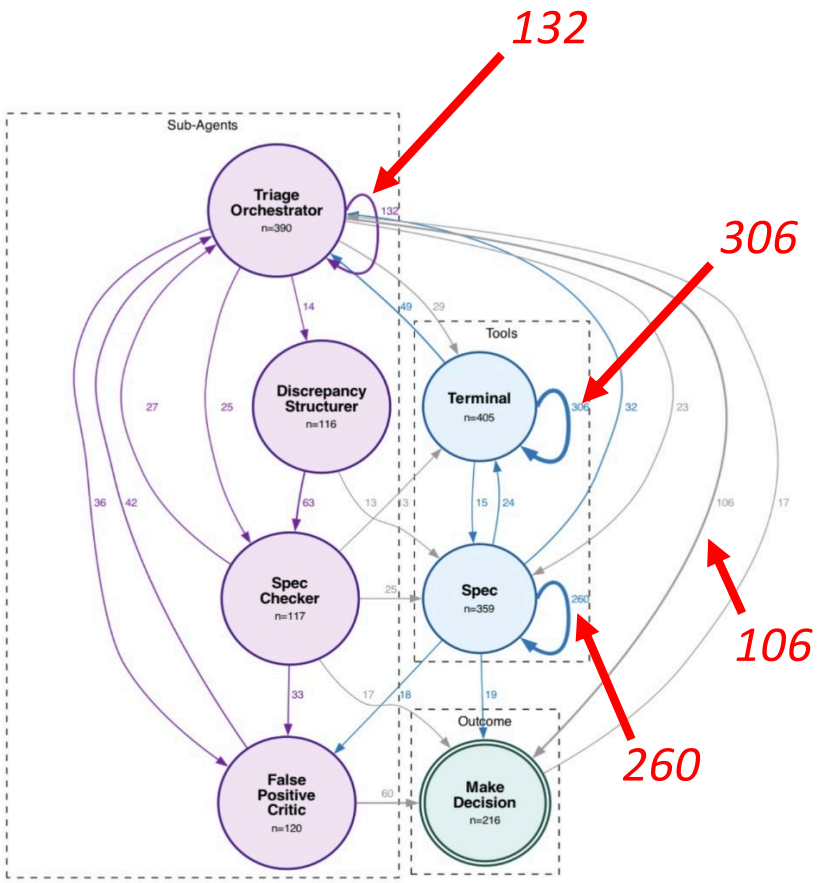



Fig. 8. Agent and tool interaction flows in the triage process (234 logged reasoning steps)

SmartOracle - An Agentic Approach:
Srinath Srinivasan
https://arxiv.org/pdf/2601.15074

# Road Map


**01** **The Bigger-is-Better Myth**
Problems with LLMs · vibe coding

**02** **State Spaces & Search**
From Tic-Tac-Toe to SE · funneling

**03** **KLAS: Knowledge-Level Agents + Tools**
Modeling trap · 1989 → 2026 · tools beat agents

**04** **EZR: Active Learning**
Sparsity at every layer · 32 samples → 90%

**05** **Why Join This Research?**
Alien code · open problems · the invitation

# What do I do? How can I help you?

papers | pre-prints | dblp

timm@ieee.org | +1-304-376-2859 | rm 3304, EB2 (map)

http://timm.fyi

So I seek **talented** grad students & **industrial partners** to find + fix the **problems in real-world AI/ML**.
**Is that you?** Maybe "yes" if **you want to be a leader in AI** (and not just another follower).

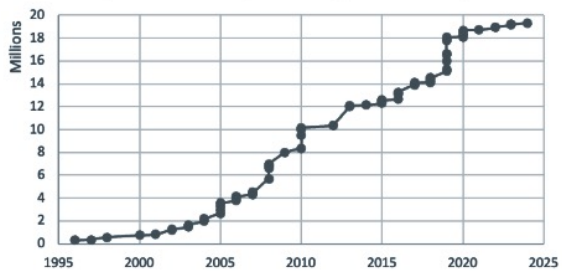
Schedule an Appointment

videos |

**Currently:**

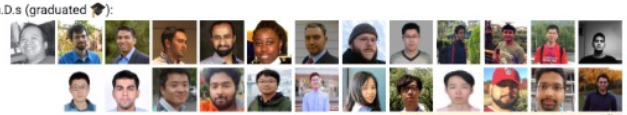
- **Teaching:** Graduate Intro to SE
- **Editor-in-chief:** Automated Software Engineering journal
- **Associate Editor:** IEEE Trans SE
- **Editorial Board:** Communications of the ACM (opinions)
- **Track co-chair:** ESEM'25 (doctoral symposium)
- **Program committees:** ICSE'26, ICSE'25, FSE'25, ASE'25, SANER'25, ESEM'25, CAIN'25, AAAI(ethics)'25

total funding
(in 2025 dollars, assuming 3% inflation)




Ph.D.s (current):

Ph.D.s (graduated 🎓):



Much thanks and good luck to y'all!. 👏
Can't wait to see what you do next! 🚀

National Science Foundation

Microsoft

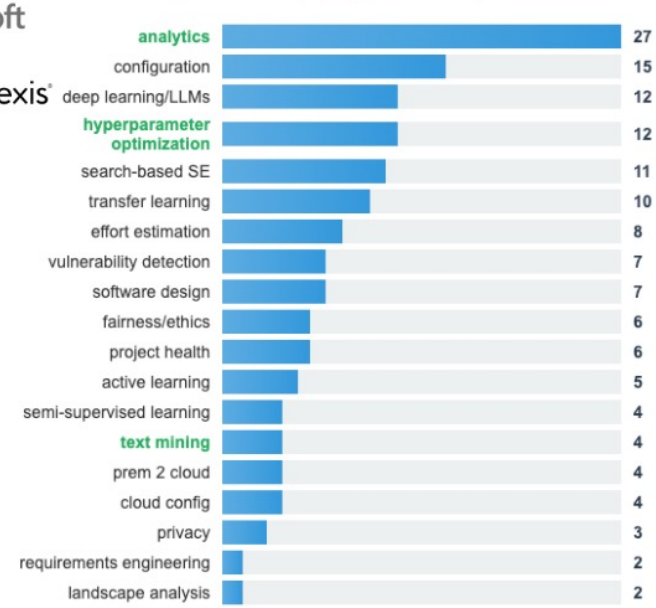
LexisNexis

National Archives

Meta

**Last decade: ~140 papers**

(estimated ~5 tags per paper, 2014-2025)




🚀 **Major Commercial Breakthrough Technologies:**

- **NASA-Proven Analytics Platform** - Mission-critical systems, space-grade reliability (27 papers)
- **Microsoft Hyperparameter Optimization** - 210X faster, enterprise validated (12 papers)
- **LexisNexis Text Mining Platform** - Legal document analysis, millions of documents
- **Cloud Configuration Systems** - Prem 2 cloud (4) + cloud config (4) solutions
- **Software Design Intelligence** - AI-driven design optimization (7 papers)
- **Project Health Analytics** - Predictive monitoring and early warning systems (6 papers)

ACT 1

# The Bigger-is-Better Myth

*Why LLMs aren't always the answer*

# Soon, will "Bigger is Better" ➔ "Smaller is Essential"?

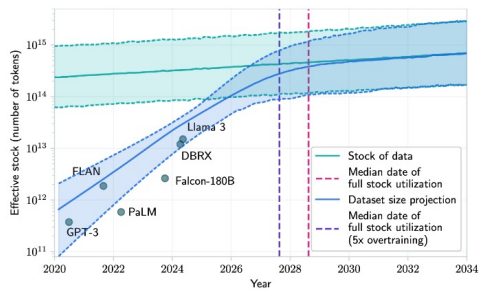
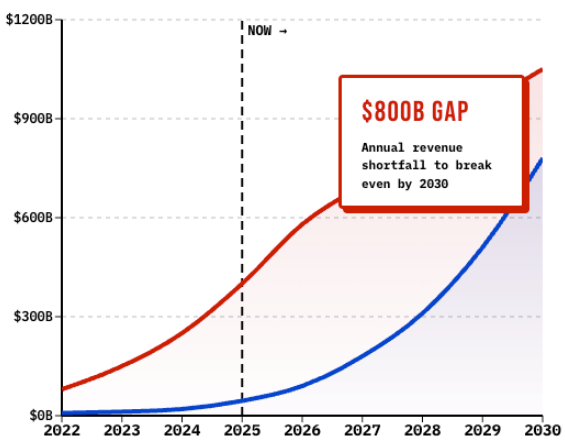
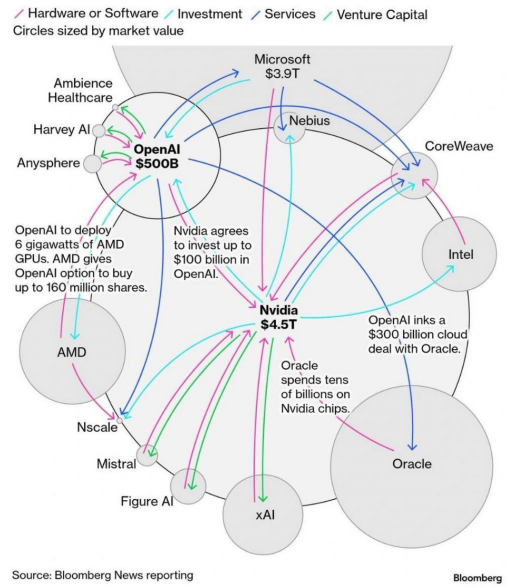



Fig. 4. Median results from the Villalobos et al. model (shown in green) estimate that by 2028, we will run out of new textual data needed to train bigger and better LLMs [75].




NEWSLETTERS • CFO DAILY

## MIT report: 95% of generative AI pilots at companies are failing

By Sheryl Estrada
Senior Writer And Author Of CFO Daily

Add us on

August 18, 2025, 6:54 AM ET

# The LLM Problem Table

Evidence from the literature

| Issue | Evidence |
|---|---|
| Cognitive debt | With generative and agentic AI accelerating how software gets built, teams can move very fast while slowly losing shared understanding of why systems are designed the way they are and how they can be safely changed [90] |
| Hallucinations | 52% of ChatGPT answers have incorrect info [38] |
| Human error | Humans overlook AI errors 39% of the time [38] |
| Code quality | 70% faulty: 29% correct, 51% partial, 20% incorrect [107] |
| Expert caution | Google developers accept only 25% of AI suggestions cite |
| Training bias | LLMs regenerate bugs from training data [37] |
| Task failure | LLMs fail at technical debt tasks [55] |
| **Additional Concerns from this document** | |
| Money | Providing student access to closed models can be expensive |
| Energy | High consumption. "Green SE" a hard constraint |
| Explanation | Subsymbolics: no causal/ explanatory reasoning |
| Verification | Students can prompt but often cannot verify output |
| Planning | "Vibe coding" skips planning, increasing testing costs |
| Ownership | Copy-pasting reduces developer job satisfaction |
| Maintenance | Lack of "under-the-hood" skills → disasters? |
| Understanding | Hard to debug/edit auto-generated code |
| **Concerns from the AI Marketplace** | |
| Bubble bursting | Market place jitters– see Table 3. |
| Subsidies ending? | LLM costs to rise 10x–100x ? |

**52%** ChatGPT answers contain incorrect info

**70%** AI-generated code is faulty

**25%** Google devs accept AI suggestions

**100x** LLM costs projected to rise

# Vibe Coding Breaks SE

Planning disappears — testing debt explodes

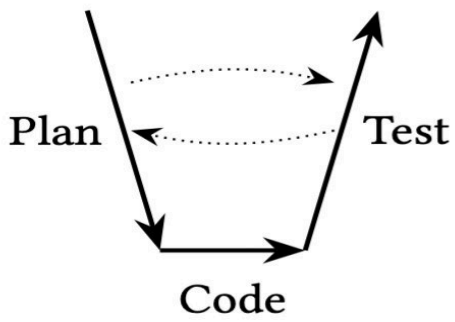



**Fig 1.a.** Current SE: Coding to gain feedback balancing planning and testing.

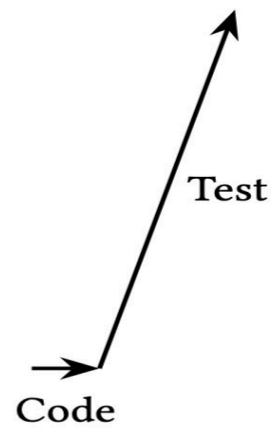



**Fig 1.b.** Vibe coding does little (or no) planning and is testing-intensive.

# Can we solve 120+ Multi-Objective Optimization Software Engineering Tasks, without LLMS?

| # Datasets | Dataset Type | File Names | Primary Objective | x/y | # Rows | Cited By |
|---|---|---|---|---|---|---|
| 25 | Specific Software Configurations | SS-A to SS-X, billing10k | Optimize software system settings | 3-88/2-3 | 197–86,059 | [24, 44–46, 49–51, 55, 57, 58] |
| 12 | PromiseTune Software Configurations | 7z, BDBC, HSQLDB, LLVM, PostgreSQL, dconvert, deeparch, exastencils, javagc, redis, storm, x264 | Software performance optimization | 9-3 [illegible]/1 | [illegible] 6,97[illegible] | [illegible] |
| 1 | Cloud | HSMGP num | Hazardous Software, Management Program data | 14/1 | 3,457 | [15, 24, 46, 57, 58] |
| 1 | Cloud | Apache AllMeasurements | Apache server performance optimization | 9/1 | 192 | [15, 24, 46, 57, 58] |
| 1 | Cloud | SQL AllMeasurements | SQL database tuning | 39/1 | 4,654 | [24, 46, 57, 58] |
| 1 | Cloud | X264 AllMeasurements | Video encoding optimization | 16/1 | 1,153 | [24, 46, 57, 58] |
| 7 | Cloud | (rs—sol—wc)* | misc configuration tasks | 3-6/1 | 196–3,840 | [24, 45, 46, 50, 57, 58] |
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| 4 | Software Process Model | XOMO (Flight, Ground, OSP) | Optimizing risk, effort, defects, and time | 27/4 | 10,001 | [11, 43–46, 58] |
| 3 | Miscellaneous | auto93, Car_price, Wine_quality | Miscellaneous | 5-38/2-5 | 205–1,600 | [24, 43, 45, 46, 57, 58] |
| 4 | Behavioral | all_players, student_dropout, HR-employeeAttrition, player_statistics | Analyze and predict behavioral patterns | 26-55/1-3 | 82–17,738 | From [1, 21, 53, 54] |
| 4 | Financial | BankChurners, home_data, Loan, Telco-Churn | Financial analysis and prediction | 19-77/2-5 | 1,460–20,000 | From [10, 18, 28, 72] |
| 3 | Human Health Data | COVID19, Life_Expectancy, hospital_Readmissions | Health-related analysis and prediction | 20-64/1-3 | 2,938–25,000 | From [19, 30, 56] |
| 2 | Reinforcement Learning | A2C_Acrobot, A2C_CartPole | Reinforcement learning tasks | 9-11/3-4 | 224–318 | |
| 5 | Sales | accessories, dress-up, Marketing_Analytics, socks, wallpaper | Sales analysis and prediction | 14-31/1-8 | 247–2,206 | From [20, 34, 60] |
| 2 | Software testing | test120, test600 | Optimize the class | 9/1 | 5,161 | |
| 127 | **Total** | | | | | |

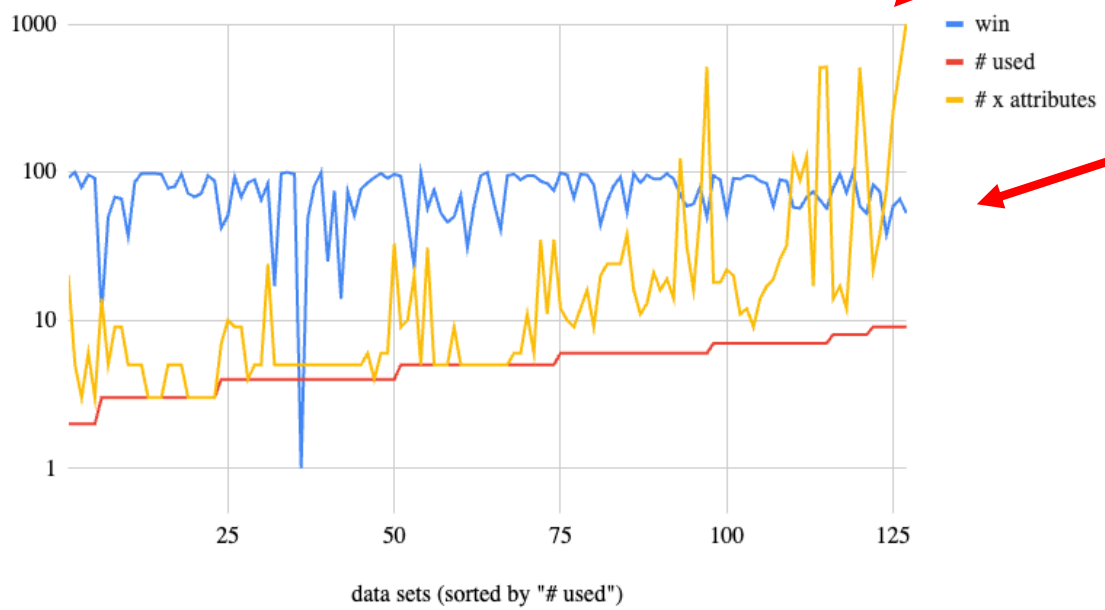
**http://github.com/timm/moot**

# MOOT= 120+ Many Multi-Objective Optimization Tasks
# Learner = EZR (see later)

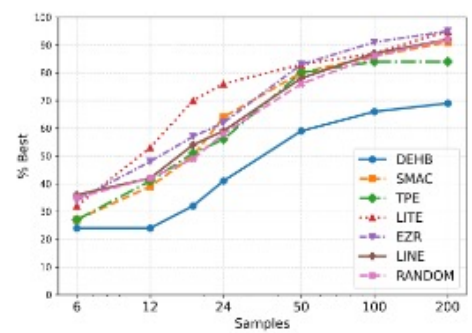
Strange tales of model complexity

Q1: Given N attributes, how many needed?
A1: Surprisingly few

Q2: How many examples to find core attributes?
A2: Surprisingly few (50)




Q3: Are models learned this way sub-optimal?
A3: Nope



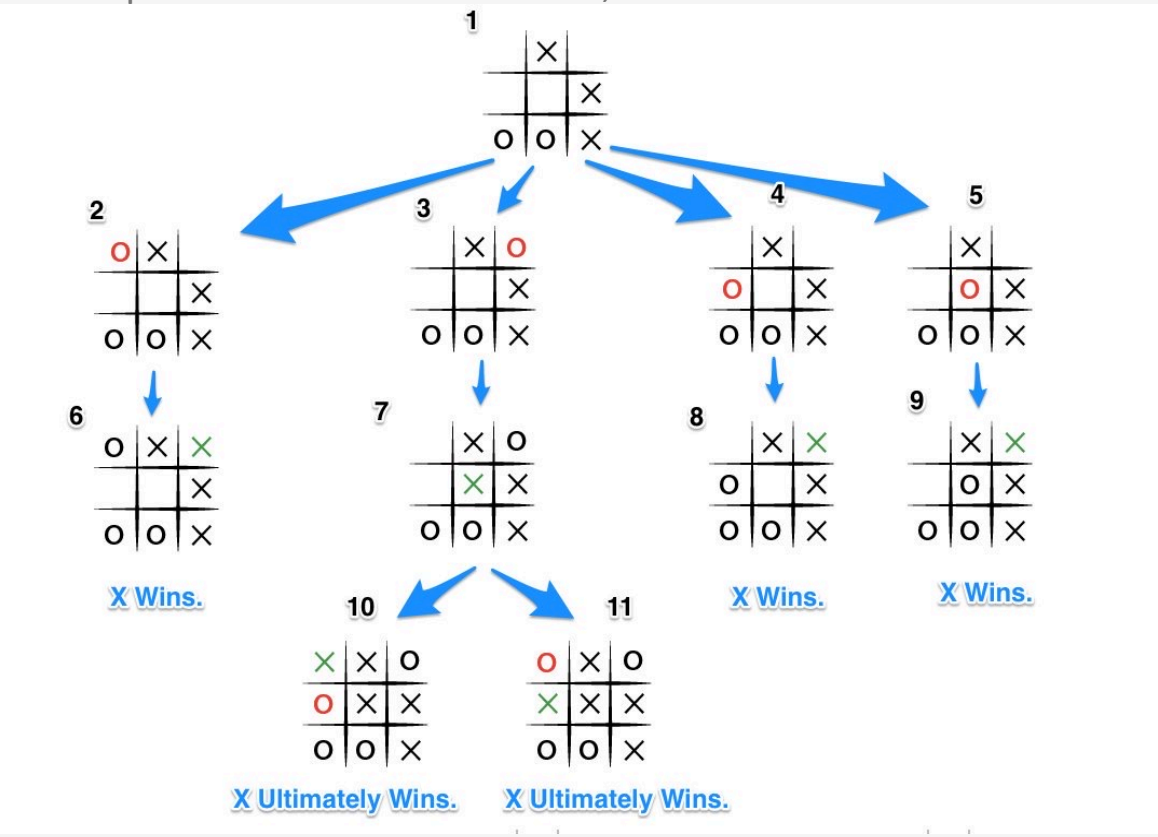
ACT 2

# State Spaces & Search

*From Tic-Tac-Toe to Software Engineering*

# Every AI Problem is a Search

State spaces: visible in Tic-Tac-Toe, infinite in SE




## Tic-Tac-Toe

~255,168 possible games.
We can see the whole tree.

## SE Decisions

Config spaces: 10^40 options.
The tree is effectively infinite.

## The Question

Find best paths without exhaustive search — how?

# The Funneling Effect

Despite complexity, software converges to few outcomes

Huge State Space (10^40 combinations)

Most paths lead nowhere useful

Behavior clusters into few patterns

A handful of key decisions matter

✓ 32 samples → 90% optimal

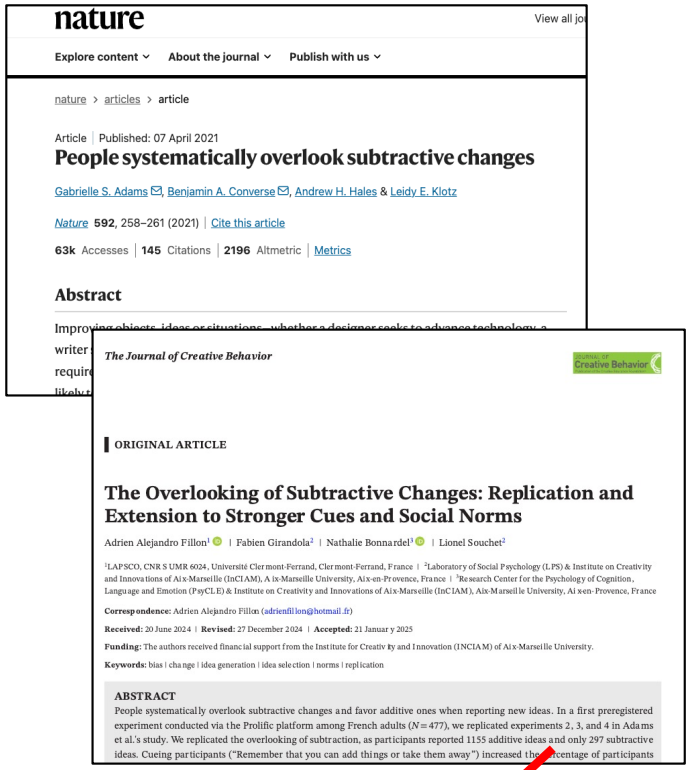
SVM+TF-IDF
beat Big AI
100× faster
better accuracy

# Q: Why aren't others reporting simplicity?

A: because they are not looking for it.

A recent survey of 229 SE papers explored LLMs for SE,

- only 5% compared their results against a simpler (non-LLM) approach.



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Article | Published: 07 April 2021

**People systematically overlook subtractive changes**

Gabrielle S. Adams, Benjamin A. Converse, Andrew H. Hales & Leidy E. Klotz

*Nature* **592**, 258–261 (2021) | Cite this article

**63k** Accesses | **145** Citations | **2196** Altmetric | Metrics

**Abstract**

Improving objects, ideas or situations—whether a designer seeks to advance technology, a writer require likely

*The Journal of Creative Behavior*

Journal of Creative Behavior

ORIGINAL ARTICLE

**The Overlooking of Subtractive Changes: Replication and Extension to Stronger Cues and Social Norms**

Adrien Alejandro Fillon[1] | Fabien Girandola[2] | Nathalie Bonnardel[3] | Lionel Souchet[2]

[1]LAPSCO, CNRS UMR 6024, Université Clermont-Ferrand, Clermont-Ferrand, France | [2]Laboratory of Social Psychology (LPS) & Institute on Creativity and Innovations of Aix-Marseille (InCIAM), Aix-Marseille University, Aix-en-Provence, France | [3]Research Center for the Psychology of Cognition, Language and Emotion (PsyCLE) & Institute on Creativity and Innovations of Aix-Marseille (InCIAM), Aix-Marseille University, Aix-en-Provence, France

**Correspondence:** Adrien Alejandro Fillon (adrienfillon@hotmail.fr)

**Received:** 20 June 2024 | **Revised:** 27 December 2024 | **Accepted:** 21 January 2025

**Funding:** The authors received financial support from the Institute for Creativity and Innovation (INCIAM) of Aix-Marseille University.

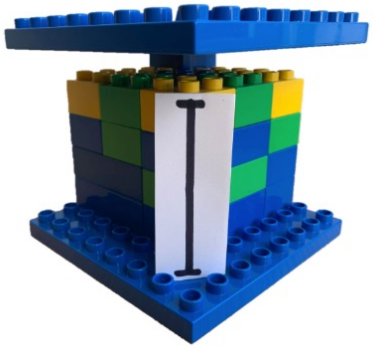
**Keywords:** bias | change | idea generation | idea selection | norms | replication

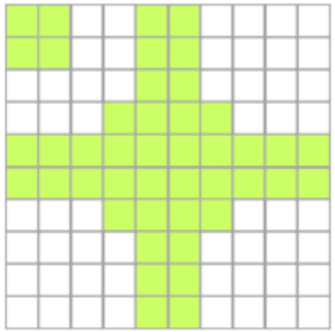
**ABSTRACT**

People systematically overlook subtractive changes and favor additive ones when reporting new ideas. In a first preregistered experiment conducted via the Prolific platform among French adults ($N = 477$), we replicated experiments 2, 3, and 4 in Adams et al.'s study. We replicated the overlooking of subtraction, as participants reported 1155 additive ideas and only 297 subtractive ideas. Cueing participants ("Remember that you can add things or take them away") increased the percentage of participants

1155 additive ideas and only 297 subtractive

Lego design (keep a block height "I" above ground)





Design a logo (make symmetrical)

ACT 3

# KLAS: Agents + Tools

*Knowledge-Level Agent Systems — what's old is new again*

# The Modeling Trap

ASP and Probabilistic Programming share a fatal assumption

| Answer Set Programming | Probabilistic Programming (Stan, Pyro) |
| --- | --- |
| Encodes system as logical rules.<br>Powerful — but requires full logical encoding of a million-line system. | Specify priors and relationships.<br>Powerful — but requires a probabilistic model T of the whole system. |

⚠️ **The Common Fatal Assumption:**

**T is cheaper to build than the software itself.**

*"In modern SE, the system IS the model. Recreating a logical shadow of it is a boil-the-ocean strategy."*

# What's Old is New

The 1989 expert system dream — now finally working

## 1989 — Newell & Yost

"Subgoals are generated whenever problem solving cannot proceed until another problem space has performed some subtask."

Architecture: Symbolic logic, hand-coded operators

Problem: Brittle, hard to scale

40 years

→

## 2026 — DeepMind

"AI agents need to decompose problems into manageable sub-components and safely delegate their completion across other AI agents."

Architecture: Same — hierarchical delegation

Engine: LLM reasoning swapped in for brittle scripts

# Agents Talk to Tools, Not Each Other

Logged reasoning steps in a real triage system

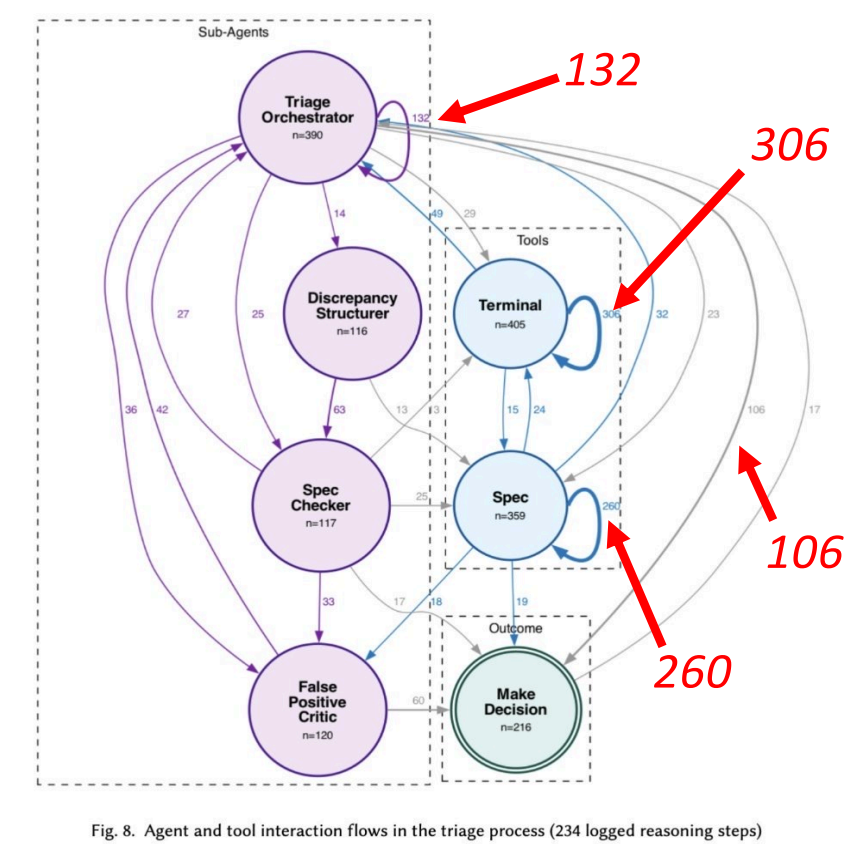



Fig. 8. Agent and tool interaction flows in the triage process (234 logged reasoning steps)

**Tools have MORE traffic than agent-to-agent calls**

**Terminal**

n=405 interactions

**Spec tool**

n=359 interactions

**Triage Orch.**

n=390 interactions

*Agents are orchestrators. Tools do the real work.*

SmartOracle - An Agentic Approach: Srinath Srinivasan https://arxiv.org/pdf/2601.15074

# Sparsity at Every Layer

A necessary condition — if software weren't sparse, we couldn't build it

**1**

### Logic / SAT

Backdoor variables: 6,750-variable problems solved by just 12 'master key' variables (Selman et al.)

**2**

### Source Code

20% of files contain 80% of bugs. Defects cluster. Power-law distributions throughout (Ostrand et al.)

**3**

### Runtime

Big-data apps process petabytes yet exercise fewer than 50 distinct execution paths (Zhang et al.)

**4**

### Design

NASA deep-space missions: optimization converges on fewer than 12% of decisions (Feather & Menzies)

*Why? Miller's Law: humans hold 7±2 chunks in working memory. Software written by humans must be sparse.*

ACT 4

# EZR: Active Learning

*Small data. Big results.*

# How EZR Works

Partial info · limited labels · exploit sparsity

```
def guess(d:Data, Any=4, Budget=50, label=lambda r:r, **kwargs) -> Data:
  rows   = shuffle(d.rows[:])
  unseen = clone(d, rows[Any:][:the.Few])
  seen   = clone(d, rows[:Any])
  seen.rows.sort(key= (Y := lambda r: disty(seen, r)))
  n      = round(sqrt(Any))
  best   = adds(seen.rows[:n], clone(d))
  rest   = adds(seen.rows[n:], clone(d))
  while unseen.n > 2 and seen.n < Budget:
    add(seen,
      add(best,
        label(
          sub(unseen,
            acquire(seen, best, rest, unseen, **kwargs)))))
    if best.n > sqrt(seen.n):
      best.rows.sort(key = Y)
      while best.n > sqrt(seen.n):
        add(rest,
          sub(best, best.rows[-1]))
  return clone(d, sorted(seen.rows, key = Y))
```

1

2

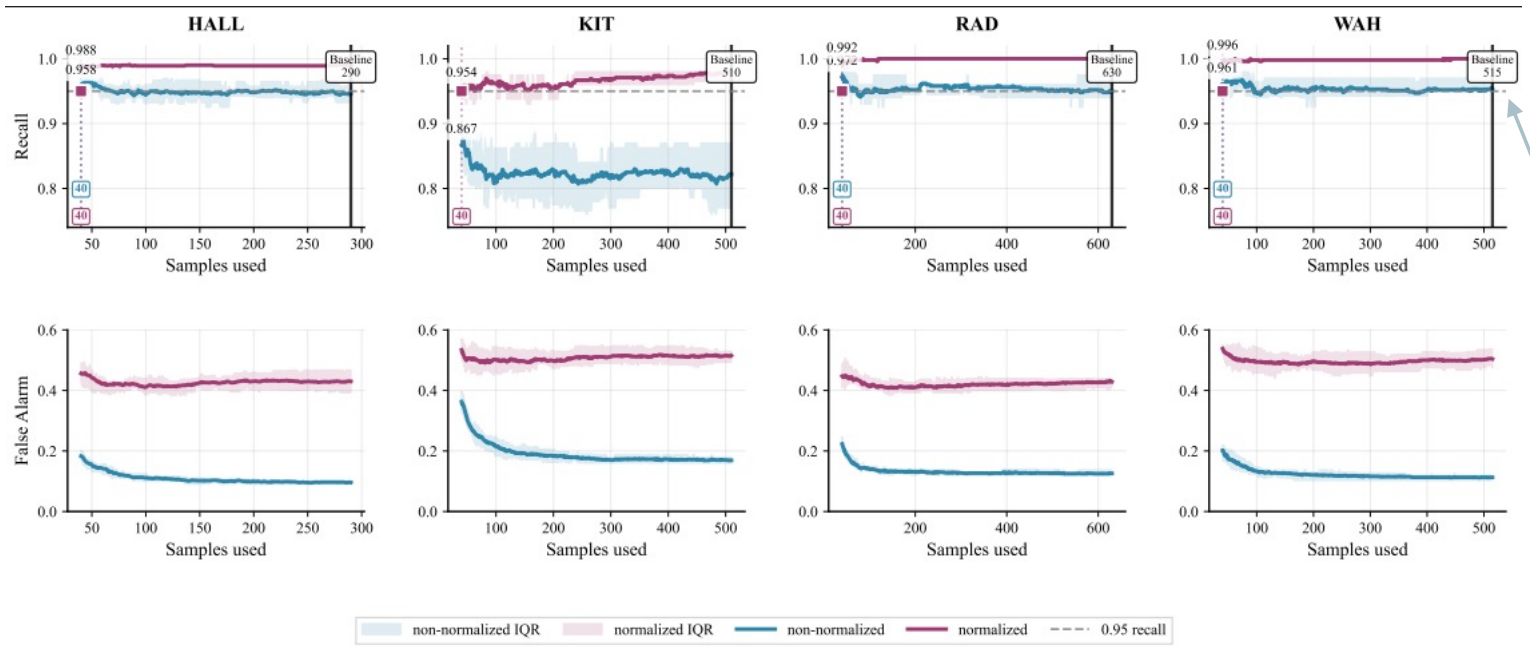
3

2’

1. **Label**
   N=4 random examples to start
2. **Split**
   √N best vs N−√N rest
3. **Classify**
   Bayes: which unlabeled looks like best?
4. **Acquire**
   Label the most informative example
5. **Loop**
   Repeat until budget exhausted
6. **Return**
   Tiny explainable regression tree

# Text mining results: find M relevant docs in N samples
# Applications: local RAGS

M/N = 104/8911 = 1.2%  M/N = 45/1502 = 3%  M/N = 48/6000 = 0.8%  M/N = 62/7002 = 0.9%




$$\frac{log(\theta_i)}{\sum \theta_i}$$

Notes:
(1) No deep learning embeddings
(2) Resource lite (to say the least)
(3) So just how easy is text mining?

Incremental Transformed Weight-normalized Complement Naive Bayes:
Training = increments TF.IDF counts in every other class
Classify = report class least likely to not belong to
How to select next example? See last slide

# MOOT= 120+ Many Multi-Objective Optimization Tasks

| # Datasets | Dataset Type | File Names | Primary Objective | x/y | # Rows | Cited By |
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| 12 | PromiseTune Software Configurations | 7z, BDBC, HSQLDB, LLVM, PostgreSQL, dconvert, deeparch, exastencils, javagc, redis, storm, x264 | Software performance optimization | 9-35/1 | 864-166,975 | [14, 15, 25, 27, 67] |
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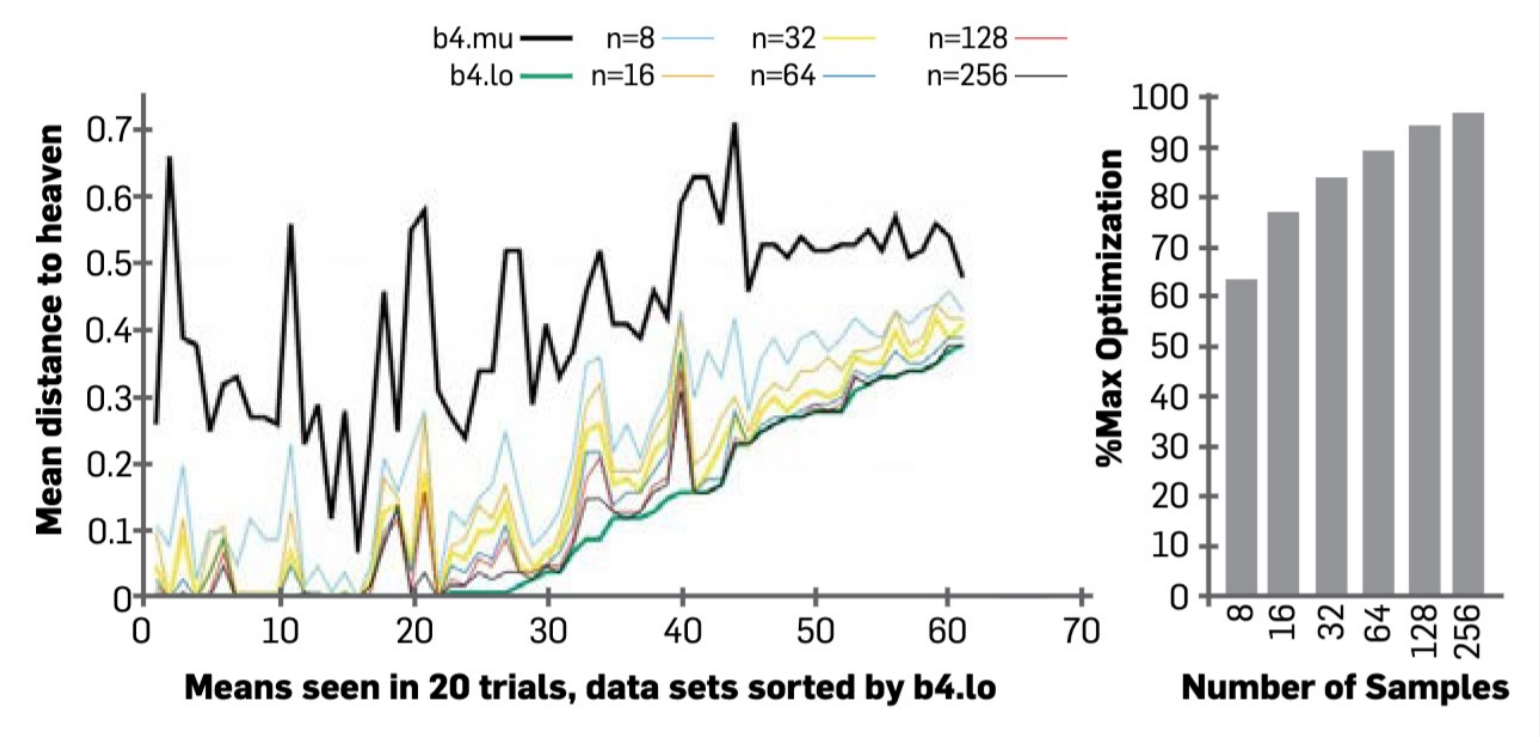
| 3 | Miscellaneous | auto93, Car_price, Wine_quality | Miscellaneous | 5-38/2-5 | 205–1,600 | [24, 43, 45, 46, 57, 58] |
| 4 | Behavioral | all_players, student_dropout, HR-employeeAttrition, player_statistics | Analyze and predict behavioral patterns | 26-55/1-3 | 82–17,738 | From [1, 21, 53, 54] |
| 4 | Financial | BankChurners, home_data, Loan, Telco-Churn | Financial analysis and prediction | 19-77/2-5 | 1,460–20,000 | From [10, 18, 28, 72] |
| 3 | Human Health Data | COVID19, Life_Expectancy, hospital_Readmissions | Health-related analysis and prediction | 20-64/1-3 | 2,938–25,000 | From [19, 30, 56] |
| 2 | Reinforcement Learning | A2C_Acrobot, A2C_CartPole | Reinforcement learning tasks | 9-11/3-4 | 224–318 | |
| 5 | Sales | accessories, dress-up, Marketing_Analytics, socks, wallpaper | Sales analysis and prediction | 14-31/1-8 | 247–2,206 | From [20, 34, 60] |
| 2 | Software testing | test120, test600 | Optimize the class | 9/1 | 5,161 | |
| 127 | **Total** | | | | | |

**http://github.com/timm/moot**

# EZR Results

63 SE optimization tasks · 20 repeated trials · standard laptop. 3 minutes



ACT 5

# Why Join This Research?

*The Invitation*

# Open Problems — Your Chance

We are not building bigger hammers. We are designing smarter architects.

### 🌍 Sustainability

Green SE: compact AI runs in minutes on a laptop, not weeks on a cluster.

### 🤝 Human-AI Partnership

EZR asks only the most informative questions. AI assists — it does not replace.

### 🔍 Explainability

Tiny regression trees. Project managers can audit and trust every decision.

### 🔬 MOOT Repository

63+ SE tasks. Open data. Real benchmarks. Join us: add your problem to MOOT.

### ⚡ Efficiency

SVM+TF-IDF beats Big AI: 100× faster, better accuracy. Do more with less.

The Open Question:

# "Alien Code"

Sparsity exists because **human cognitive limits** force it.

Humans obey Miller's Law: 7±2 chunks of working memory.
So the code humans write is necessarily regular and sparse.

**LLMs face no such constraint.** They can generate code with dense, high-dimensional dependencies ($|A'| \gg 10$).

Will AI code become **"unher-dable"?** Do we need **"Sparsity Constraints"** as a safety requirement for generative AI?

**This is YOUR research problem. Nobody has the answer yet.**

# Come Rediscover the Future

EZR · MOOT · KLAS · Active Learning

timm@ieee.org · NC State · github.com/timm

*"Intelligence requires more than just size."*

— The Case for Compact AI, CACM 2025